Intelligent 180° Low-Light Scanning System

Engineering Portfolio Project #1

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Grade: 10

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1. Project Overview

I wanted to build a system that could automatically scan an area and detect places with low light. Many devices around us, such as automatic lights, security systems, and robots, use sensors to collect information and make decisions. This project allowed me to learn how a sensor, a motor, and a microcontroller can work together to complete a simple task.

The system uses an Arduino Uno, a photoresistor (light sensor), a servo motor, and an LED. The servo slowly rotates the light sensor from 0° to 180°, allowing it to check the light level in different directions. If the sensor detects that the light level is below a preset value, the Arduino turns on the LED to show that a dark area has been found. The system continues scanning automatically without any user input.

This project helped me learn about electronics, programming, and how different components can work together to create an automated system.

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2. The Problem I Wanted to Solve

Sometimes it is useful for a device to automatically detect changes in its surroundings instead of relying on a person to check them. I wanted to build a simple system that could scan an area, measure the amount of light, and react whenever it found a darker location.

Although this project is simple, the same basic idea is used in many real-world technologies such as automatic lighting, robots, environmental monitoring, and security systems.

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3. Project Goals

Before starting the project, I set a few goals:

- Build a system that automatically scans a 180° area.
- Measure the light level using a photoresistor.
- Turn on an LED whenever the light level is below a selected threshold.
- Continue scanning without stopping.
- Learn how sensors, motors, and programming work together in one system.

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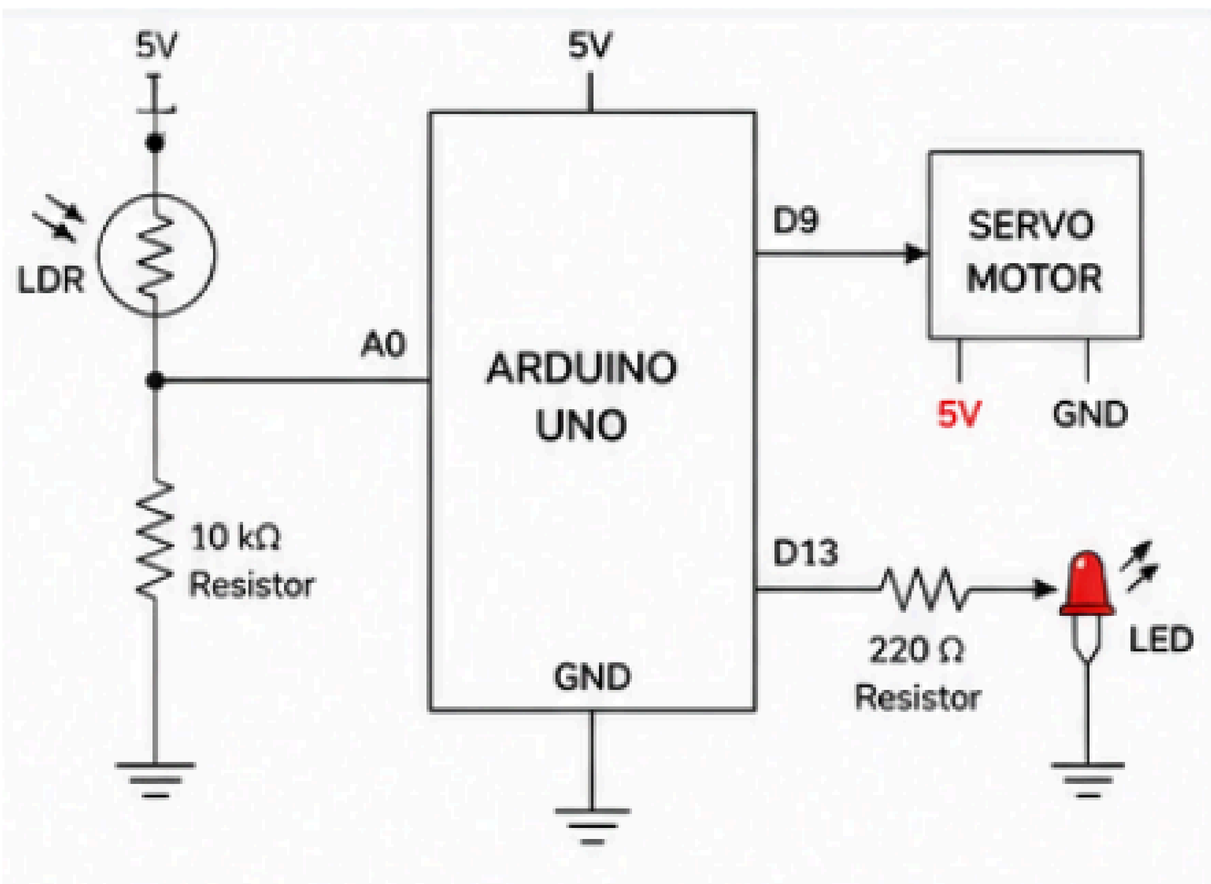
4. System Architecture



5. How My System Works

The Arduino controls the entire project.

First, the servo motor rotates the light sensor from one side to the other, scanning an area from 0° to 180°. At each position, the photoresistor measures the amount of light and sends that information to the Arduino.

The Arduino compares the sensor reading with a preset threshold. If the light level is low, it turns on the LED. If there is enough light, the LED stays off.

When the servo reaches 180°, it rotates back to 0° and repeats the process continuously.

This creates an automatic scanning system that can constantly monitor the surrounding light levels.

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6. Components Used and Why I Chose Them

Component	**Why I Used It**
Arduino Uno	The Arduino is the brain of the project. It reads the light sensor and controls both the servo motor and the LED. I chose it because it is easy to program and works well for beginner engineering projects.
Servo Motor	The servo rotates the light sensor from 0° to 180°. I chose a servo because it can move to specific angles very accurately.
Photoresistor (LDR)	The photoresistor measures how much light is present. Its resistance changes depending on the amount of light it receives.
LED	The LED gives a simple visual signal whenever a dark area is detected.
220 Ω Resistor	This resistor limits the electrical current going through the LED so it is not damaged.
10 kΩ Resistor	This resistor works with the photoresistor to create a voltage divider, allowing the Arduino to measure changes in light level.
Breadboard	The breadboard allowed me to build and test the circuit without soldering any parts together.
Jumper Wires	The jumper wires connected all of the electronic components together.

7.Code

```
#include <Servo.h>

int sensor = A0;

int LV;

int servoPin = 9;

int dt = 400;

int led = 5;

Servo MyServo;


void setup() {

  Serial.begin(9600);

  MyServo.attach(servoPin);

  pinMode(led, OUTPUT);

}

void loop() {

  // Sweep 0 → 180

  for (int pos = 0; pos <= 180; pos += 20) {

    MyServo.write(pos);


    LV = analogRead(sensor);

    Serial.println(LV);


    if (LV <= 300) {

      analogWrite(led, 255);   // Dark = fully on

    }

    else if (LV <= 500) {

      analogWrite(led, 100);   // Slightly dim = dim LED

    }
```

```
    else {
      analogWrite(led, 0);     // Bright = off
    }
          delay(dt);
  }
          // Sweep 180 → 0
  for (int pos = 180; pos >= 0; pos -= 20) {
    MyServo.write(pos);

    LV = analogRead(sensor);
    Serial.println(LV);

    if (LV <= 300) {
      analogWrite(led, 255);
    }
    else if (LV <= 500) {
      analogWrite(led, 100);
    }
    else {
      analogWrite(led, 0);
    }

    delay(dt);
  }
}
```

7. Skills I Learned

While building this project, I learned how to:

- Build a basic electronic circuit using a breadboard.
- Program an Arduino using C++.
- Read values from an analog sensor.
- Control the position of a servo motor.
- Use an LED as an output device.
- Test and troubleshoot wiring problems.
- Adjust the light threshold so the system responded correctly.

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8. Challenges I Faced

Like most projects, this one did not work perfectly the first time.

One challenge was finding the right light threshold. If the threshold was too high, the LED stayed on most of the time. If it was too low, the LED rarely turned on. I tested different values until I found one that worked well.

Another challenge was ensuring all the wires were connected correctly. A loose connection caused the servo motor to stop working during testing. By checking the wiring carefully, I was able to find the problem and fix it.

These challenges taught me the importance of testing each part of the system before putting everything together.

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9. What I Learned

This project helped me understand that even simple electronic systems require hardware and software to work together.

I learned how a sensor collects information, how a microcontroller processes that information, and how an output device responds based on programmed instructions.

Building this project also showed me that engineering is not only about writing code or connecting wires. It involves planning, testing, solving problems, and improving the design when something does not work as expected.

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10. Future Improvements

If I continue developing this project, I would like to add several improvements:

- Replace the LED with a small display that shows the light sensor reading.
- Save the light measurements so I can compare data from different locations.
- Design a printed circuit board (PCB) instead of using a breadboard.
- Add a buzzer or another indicator when a dark area is detected.
- Use a more accurate digital light sensor to improve the measurements.

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11. Reflection

This was my first project to combine electronics, programming, and mechanical movement into a single system.

I enjoyed seeing how individual components worked together to solve a problem. The project also helped me become more confident using Arduino and introduced me to the basics of embedded systems.

Working on this project made me more interested in Electrical Engineering because it showed me how sensors, programming, and electronic circuits can be combined to build systems that interact with the real world. I look forward to building more advanced projects and continuing to improve my engineering skills.

Project Video